

CSR Scholarship Management System

Complete Technical Design Document

Student Portal • Admin Console • CSR Partner Approval • Fund Disbursement

Version 2.0

SQL Server

12 DB Tables

FAANG-Level Design

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System Overview & User Roles

What the system does, who uses it, and guiding principles

What is this System?

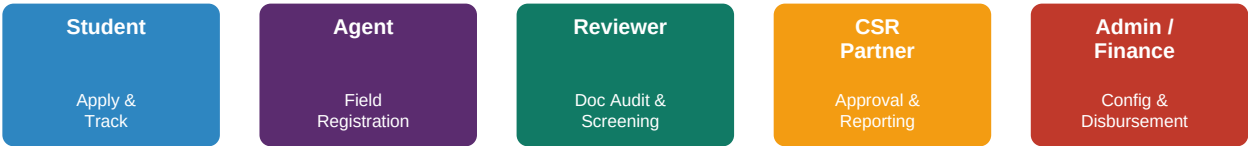
The CSR Scholarship Management System is an enterprise-grade digital platform that handles the complete end-to-end lifecycle of scholarship applications — from student self-registration and eligibility-based auto-matching, through multi-stage review (document audit, background check, screening), CSR partner approval, and finally fund disbursement directly to students or to their educational institutions. Every decision is logged for full auditability.

Guiding Principles

Transparency	Students always see exactly where their application stands in the pipeline.
Auditability	Every approval, rejection, and payment is logged with who, what, and when.
Data Privacy	Sensitive personal and economic data is access-controlled and encrypted at rest.
Configurability	Eligibility criteria and scholarship rules are configurable by admin — not hard-coded.
Separation of Duties	Review, screening, partner approval, and disbursement are enforced as distinct roles.

User Roles

5 User Roles — Each with Dedicated Dashboard



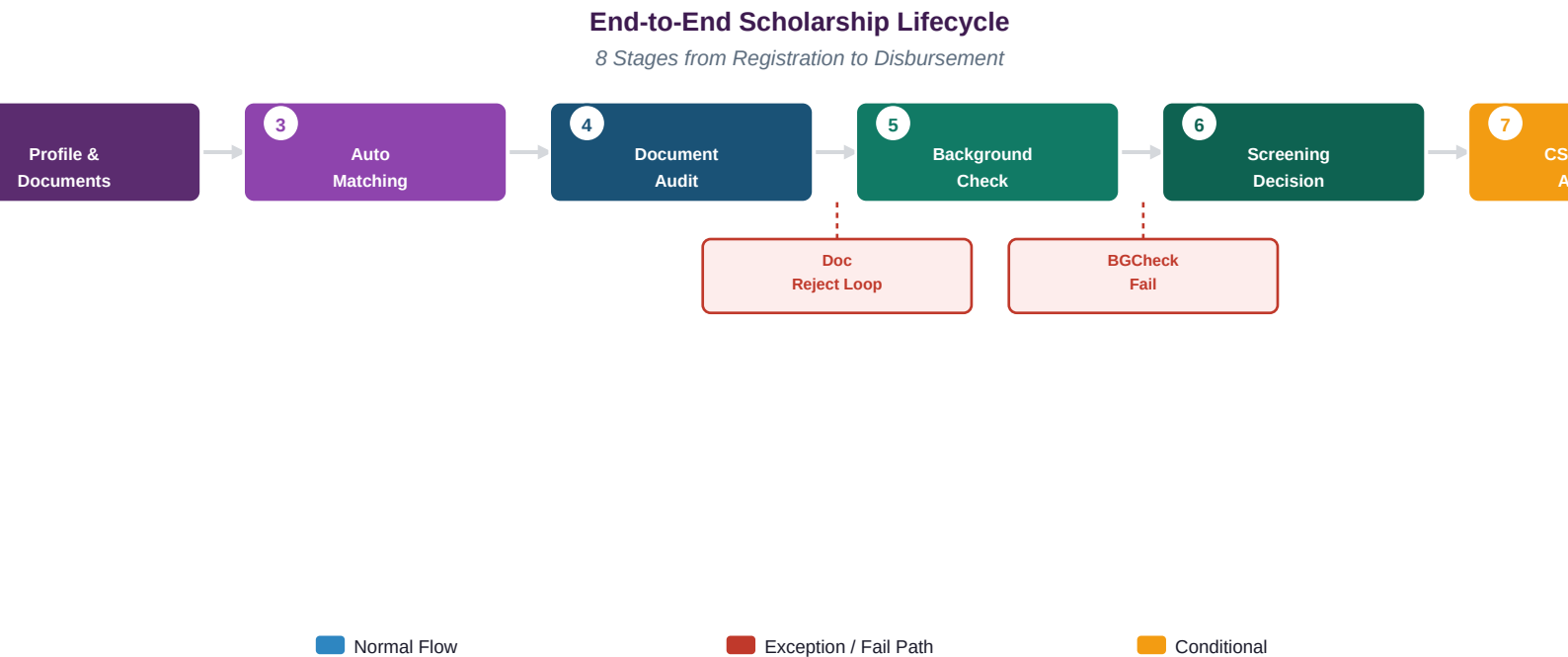
Student	Self-register, fill profile, upload documents, apply for matching scholarships, track status.
Agent	Field representative. Registers students on their behalf. Submits batch applications.
Document Reviewer	Audits uploaded documents. Flags missing or rejected items. Triggers re-upload requests.
Background Check Officer	Conducts verification checks. Records pass/fail results per case.
Screening Officer	Makes final acceptance decision based on audit + background check results.
CSR Partner	Approves or declines shortlisted students. Views fund utilization dashboard.
Admin	Configures eligibility rules, manages sponsors, oversees all stages, can override.

Finance Officer

Executes fund disbursement. Maker-checker workflow for payment approval.

End-to-End Lifecycle Workflow

Visual flow from student registration to fund disbursement



Stage Summary

Stage 1 — Student Registration	Student creates account (email/phone OTP), verifies identity, sets up login.
Stage 2 — Profile & Documents	Fills personal details, family income, education info. Uploads all required documents.
Stage 3 — Auto-Matching	System automatically matches student profile against active scholarship eligibility rules. Ineligible applications flagged immediately with reason.
Stage 4 — Document Audit	Reviewer checks each document against checklist. Rejected docs trigger re-upload loop. Only moves forward when all docs verified.
Stage 5 — Background Check	Background Check Officer records verification findings. Pass moves to screening. Fail routes back with mandatory reason.
Stage 6 — Screening Decision	Screening Officer makes final accept/reject decision. Cannot be same person as document reviewer (separation of duties).
Stage 7 — CSR Partner Approval	Shortlisted students sent to CSR partner for acceptance. Partner can approve, decline, or defer. Declined students can be re-pooled.
Stage 8 — Fund Disbursement	Finance team disburses funds to student bank account or institution account. Maker-checker workflow. Receipt generated. Student notified.

Exception & Decision Points

Eligibility Fail at Auto-Match: Application flagged ineligible; student notified with specific reason.

Document Rejected at Audit: Checklist item reopened; student re-uploads; audit resumes from that item.

Background Check Fail: Case routed to screening for rejection decision with mandatory documented reason.

Screening Reject: Student notified with reason; case closed (appeal path in future phase).

Partner Decline: Case returned to admin; candidate may be re-pooled for alternate scholarship.

Disbursement Failure: Finance officer retries or corrects beneficiary details; status stays Initiated until resolved.

Database Design — SQL Server

All 12 tables with columns, types, constraints, and relationships

Database: Microsoft SQL Server | **Total Tables:** 12 | **Convention:** All PKs are IDENTITY(1,1). All tables have CreatedAt DATETIME2 DEFAULT GETDATE().

Database Entity Relationship Overview
12 Tables — Applications at the Center

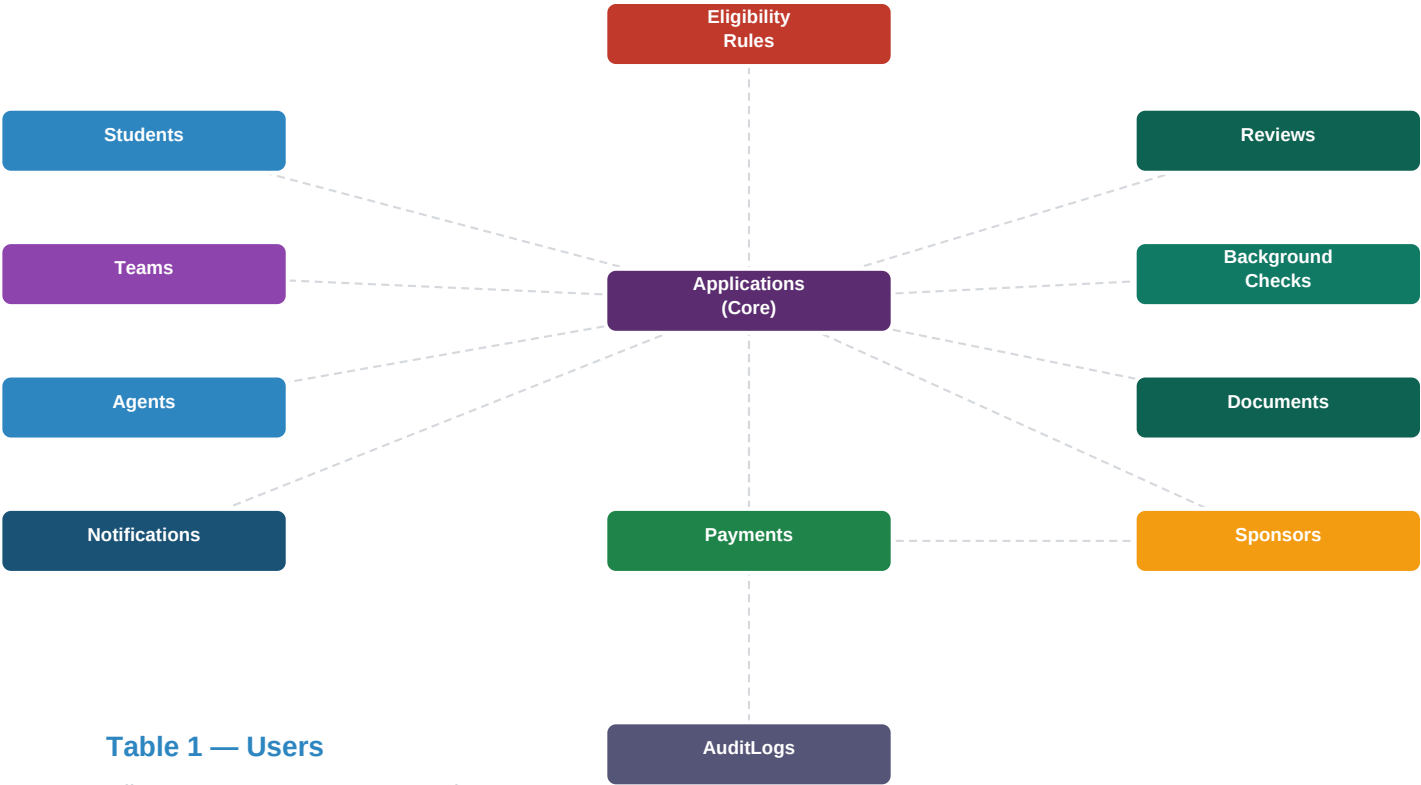


Table 1 — Users

All system users across every role.

```
1 CREATE TABLE Users (  
2   UserID INT IDENTITY(1,1) PRIMARY KEY,  
3   FullName NVARCHAR(150) NOT NULL,  
4   Email NVARCHAR(200) NOT NULL UNIQUE,  
5   Phone NVARCHAR(20),  
6   PasswordHash NVARCHAR(255) NOT NULL,  
7   Role NVARCHAR(30) NOT NULL  
8   CHECK (Role IN ('Student', 'Agent', 'DocReviewer', 'BGCheckOfficer',  
9     'ScreeningOfficer', 'CSRPartner', 'Admin', 'Finance')),  
10  AgentCode NVARCHAR(20) UNIQUE,  
11  SponsorID INT REFERENCES Sponsors(SponsorID),  
12  IsActive BIT DEFAULT 1,  
13  CreatedAt DATETIME2 DEFAULT GETDATE(),  
14  UpdatedAt DATETIME2 DEFAULT GETDATE()  
15 );
```

8 distinct roles enforced via CHECK constraint. Separation of duties is enforced in application logic — same user cannot be DocReviewer and ScreeningOfficer on the same case.

Table 2 — Students

Extended student profile — personal, family, and education details.

```
1 CREATE TABLE Students (  
2 StudentID INT IDENTITY(1,1) PRIMARY KEY,  
3 UserID INT NOT NULL REFERENCES Users(UserID),  
4 AadharNumber NVARCHAR(20),  
5 DOB DATE,  
6 Gender NVARCHAR(10),  
7 Category NVARCHAR(50), -- General/OBC/SC/ST  
8 Address NVARCHAR(500),  
9 City NVARCHAR(100),  
10 State NVARCHAR(100),  
11 Pincode NVARCHAR(10),  
12 AnnualFamilyIncome DECIMAL(15,2),  
13 FamilySize INT,  
14 Course NVARCHAR(200),  
15 InstitutionID INT REFERENCES Institutions(InstitutionID),  
16 EnrollmentYear INT,  
17 BankAccountNo NVARCHAR(50),  
18 BankIFSC NVARCHAR(20),  
19 BankName NVARCHAR(100),  
20 CreatedAt DATETIME2 DEFAULT GETDATE()  
21 );
```

InstitutionID links to the Institutions master table — needed for institution-based disbursement. **Category** field supports diversity/coverage reporting.

Table 3 — Applications

Core record tracking the full application lifecycle.

```
1 CREATE TABLE Applications (  
2 ApplicationID INT IDENTITY(1,1) PRIMARY KEY,  
3 StudentID INT NOT NULL REFERENCES Students(StudentID),  
4 AgentID INT REFERENCES Agents(AgentID),  
5 ScholarshipID INT NOT NULL REFERENCES Scholarships(ScholarshipID),  
6 SubmissionDate DATETIME2 DEFAULT GETDATE(),  
7 Status NVARCHAR(40) DEFAULT 'Draft'  
8 CHECK (Status IN ('Draft', 'Submitted', 'AutoMatched', 'EligibilityFailed',  
9 'DocAuditInProgress', 'DocAuditComplete', 'BGCheckInProgress',  
10 'BGCheckComplete', 'ScreeningPending', 'ScreeningApproved',  
11 'ScreeningRejected', 'CSRPending', 'CSRApproved', 'CSRDeclined',  
12 'PaymentPending', 'PaymentInitiated', 'PaymentCompleted', 'Cancelled')),  
13 AssignedDocReviewer INT REFERENCES Users(UserID),  
14 AssignedBGOfficer INT REFERENCES Users(UserID),
```



```

15 AssignedScreeners INT REFERENCES Users(UserID),
16 ScholarshipAmount DECIMAL(10,2),
17 SponsorID INT REFERENCES Sponsors(SponsorID),
18 AdminApprovedBy INT REFERENCES Users(UserID),
19 AdminApprovedAt DATETIME2,
20 Notes NVARCHAR(MAX),
21 CreatedAt DATETIME2 DEFAULT GETDATE()
22 );

```

18 status values cover every stage of the lifecycle. Separate assigned columns for each role enforces separation of duties at the database level.

Table 4 — EligibilityRules

Admin-configurable eligibility criteria per scholarship.

```

1 CREATE TABLE EligibilityRules (
2 RuleID INT IDENTITY(1,1) PRIMARY KEY,
3 ScholarshipID INT NOT NULL REFERENCES Scholarships(ScholarshipID),
4 RuleType NVARCHAR(50) NOT NULL
5 CHECK (RuleType IN ('MaxAnnualIncome','MinAge','MaxAge',
6 'Gender','Category','State','Course',
7 'MinGrade','InstitutionType')),
8 Operator NVARCHAR(10), -- 'LT','GT','EQ','IN','BETWEEN'
9 ValueMin NVARCHAR(200),
10 ValueMax NVARCHAR(200),
11 ValueList NVARCHAR(MAX), -- JSON array for IN operator
12 IsRequired BIT DEFAULT 1,
13 CreatedAt DATETIME2 DEFAULT GETDATE()
14 );

```

Fully configurable — admin creates rules via UI, no code change needed. Auto-matching engine reads these rules at runtime to evaluate each student profile.

Table 5 — DocumentChecklist

Required documents per scholarship with audit tracking.

```

1 CREATE TABLE DocumentChecklist (
2 ChecklistID INT IDENTITY(1,1) PRIMARY KEY,
3 ApplicationID INT NOT NULL REFERENCES Applications(ApplicationID),
4 DocumentType NVARCHAR(50) NOT NULL
5 CHECK (DocumentType IN ('Aadhar','PAN','IncomeCertificate',
6 'CasteCertificate','EducationProof','BankProof','PhotoID','Other')),
7 FileURL NVARCHAR(500),
8 UploadedAt DATETIME2,
9 ReviewedBy INT REFERENCES Users(UserID),
10 ReviewedAt DATETIME2,
11 Status NVARCHAR(20) DEFAULT 'Pending'
12 CHECK (Status IN ('Pending','Uploaded','Verified','Rejected','ReUploadRequested')),
13 RejectionReason NVARCHAR(500),

```

```

14 ReUploadCount INT DEFAULT 0,
15 CreatedAt DATETIME2 DEFAULT GETDATE()
16 );

```

ReUploadCount tracks how many times a document was rejected and re-uploaded. **RejectionReason** is shown to student so they know exactly what to fix.

Table 6 — BackgroundChecks

Stores background verification results per application.

```

1 CREATE TABLE BackgroundChecks (
2   CheckID INT IDENTITY(1,1) PRIMARY KEY,
3   ApplicationID INT NOT NULL REFERENCES Applications(ApplicationID),
4   OfficerID INT NOT NULL REFERENCES Users(UserID),
5   CheckType NVARCHAR(50), -- 'Identity','Address','IncomeVerification'
6   Result NVARCHAR(20)
7   CHECK (Result IN ('Pass','Fail','Inconclusive','Pending')),
8   Notes NVARCHAR(MAX),
9   EvidenceURL NVARCHAR(500),
10  CompletedAt DATETIME2,
11  CreatedAt DATETIME2 DEFAULT GETDATE()
12 );

```

Each application can have **multiple check records** for different check types. Fail result routes the case back to screening with mandatory notes.

Remaining Tables

Table 7 — Scholarships	ScholarshipID, Name, Description, SponsorID, TotalBudget, PerStudentAmount, ApplicationWindow (open/close dates), Status, MaxApplicants.
Table 8 — Institutions	InstitutionID, Name, Type (School/College/University), District, State, BankAccountNo, BankIFSC, IsVerified. Needed for institution disbursement.
Table 9 — Sponsors	SponsorID, SponsorName, ContactPerson, Email, TotalFund, FundAllocated, FundUtilized, ApprovalPolicy (Auto/Manual), Status.
Table 10 — Payments	PaymentID, ApplicationID, StudentID or InstitutionID, SponsorID, Amount, PaymentType (Direct/Institution), Status (Pending/Initiated/Completed/Failed), ReferenceNo, MakerID, CheckerID, CreatedAt.
Table 11 — AuditLogs	LogID, UserID, Action, EntityType, EntityID, OldValue (JSON), NewValue (JSON), IPAddress, CreatedAt. Every key action logged.
Table 12 — Notifications	NotificationID, UserID, Type, Channel (Email/SMS/InApp), Message, IsSent, SentAt, RetryCount, CreatedAt.

Eligibility Configuration & Auto-Matching

How admin sets rules and system auto-matches student profiles

Eligibility Configuration (Admin)

Admin creates scholarship-specific eligibility rules via the UI — no code changes needed. Each rule defines a criterion type, operator, and value. The system evaluates these rules automatically when a student submits their profile.

Example Rules for a Scholarship:

Rule 1: MaxAnnualIncome < 3,00,000 (family income below 3 lakhs)

Rule 2: Category IN ['SC','ST','OBC'] (reserved category)

Rule 3: State = 'Maharashtra' (geography filter)

Rule 4: Course IN ['Engineering','Medicine','Law'] (course filter)

Rule 5: MinAge >= 17, MaxAge <= 25 (age band)

Admin can add, edit, or deactivate rules without touching any code.

Auto-Matching Engine Logic

Auto-Matching Engine — Node.js Backend Logic

```
1 // autoMatcher.js — runs when student submits profile
2 async function matchStudentToScholarships(studentID) {
3   const student = await getStudentProfile(studentID);
4   const scholarships = await getActiveScholarships();
5
6   for (const scholarship of scholarships) {
7     const rules = await getEligibilityRules(scholarship.ScholarshipID);
8     let eligible = true;
9     let failReasons = [];
10
11    for (const rule of rules) {
12      const passed = evaluateRule(student, rule);
13      if (!passed) {
14        eligible = false;
15        failReasons.push(rule.RuleType + ': criteria not met');
16      }
17    }
18
19    if (eligible) {
20      // Create application — Status = 'AutoMatched'
21      await createApplication(studentID, scholarship.ScholarshipID);
22    } else {
23      // Log why student was not matched — shown on their dashboard
24      await logIneligibility(studentID, scholarship.ScholarshipID, failReasons);
25    }
26  }
27 }
```

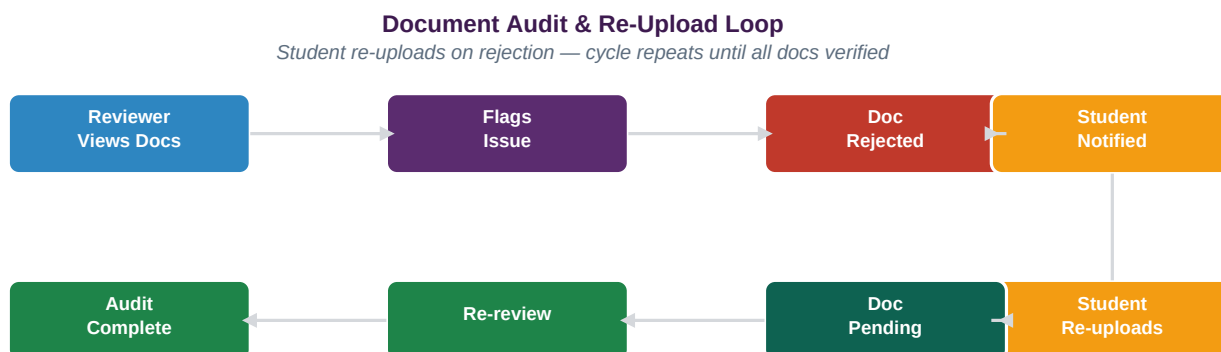
26 }

27 }

Chapter 5

Document Audit & Re-Upload Loop

How document review and rejection cycle works



Document Audit Process

Step 1 — Reviewer Opens Checklist	Doc Reviewer sees all required documents for the application. Each item shows: Pending / Uploaded / Verified / Rejected / Re-upload Requested.
Step 2 — Review Each Document	Reviewer opens each document. Checks for clarity, correctness, and validity (not expired, correct format).
Step 3 — Verify or Reject	Verify: marks doc as Verified. Reject: must enter specific rejection reason (e.g. 'Income certificate expired', 'Aadhar photo unclear').
Step 4 — Student Notified	On rejection, system sends Email + SMS to student with exact reason and deadline for re-upload (default 7 days).
Step 5 — Student Re-uploads	Student logs in, sees flagged document with rejection reason, uploads corrected version.
Step 6 — Audit Resumes	Document status resets to Pending. Reviewer is notified. Cycle repeats until document is Verified. ReUploadCount incremented each cycle.
Step 7 — Audit Complete	Once ALL documents in checklist are Verified, application status auto-updates to DocAuditComplete and moves to Background Check stage.

Background Check Module

Dedicated verification stage before screening

Background Check Workflow

After document audit is complete, the case is assigned to a Background Check Officer. This is a separate role from the Document Reviewer — enforcing separation of duties. The officer conducts verification and records structured results.

Assigned to BG Officer	System auto-assigns to available Background Check Officer or admin manually assigns. Officer notified via email + in-app.
Conduct Verification	Officer verifies identity, address, and income claim. May contact student, institution, or use third-party verification.
Record Result	Records Pass / Fail / Inconclusive per check type. Can attach evidence file (EvidenceURL in BackgroundChecks table).
Pass	Application status moves to BGCheckComplete. Case routed to Screening Officer.
Fail	Case routed back with mandatory failure notes. Screening Officer must make rejection decision — system does not auto-reject.
Inconclusive	Officer can request more time or escalate to admin for decision.

Screening Decision & CSR Partner Approval

Final acceptance and sponsor confirmation workflow

Screening Decision

Who does it: Screening Officer (cannot be same person as Doc Reviewer on same case)

Input: Completed document audit + background check results

Outcome: Screening Approved → moves to CSR Partner | Screening Rejected → student notified with reason

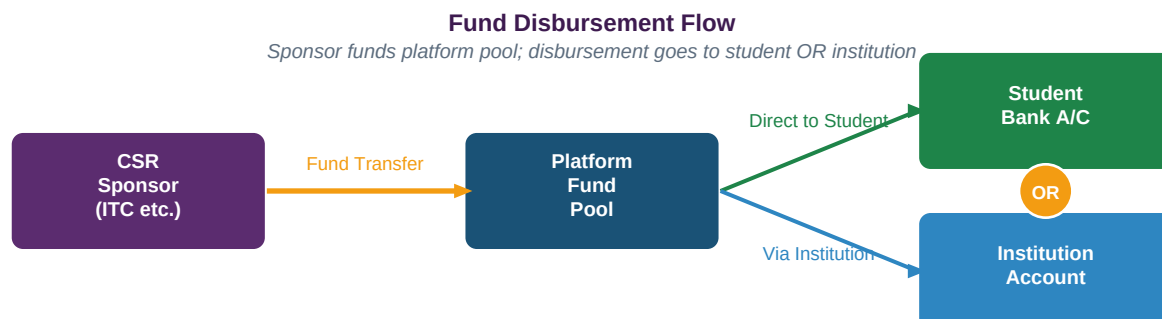
CSR Partner Approval Workflow

Shortlist Created	Admin creates shortlist of ScreeningApproved applications for a specific sponsor. Shortlist locked before sending.
Sent to CSR Partner	Partner gets email with dashboard link. Can see: student name, college, scholarship amount, eligibility summary, document verification status.
Partner Reviews	Partner reviews each student on their dashboard. Can filter by region, category, college type.
Approve	Status → CSRApproved. Student moves to payment queue. Partner cannot see bank details (data privacy).
Decline	Status → CSRDeclined. Case returned to admin. Admin can re-pool student for alternate scholarship if budget allows.
Defer	Partner can defer and review later. SLA clock tracks time taken.

Chapter 8

Fund Disbursement — Student & Institution

Payment workflow, maker-checker, and reconciliation



Two Disbursement Modes

Direct to Student	Fund transferred to student's verified bank account (from Students table — BankAccountNo + IFSC). Used when student is the primary beneficiary.
To Institution	Fund transferred to school or college bank account (from Institutions table). Used for tuition fee payment directly to institution. Student does not receive cash.

Maker-Checker Payment Workflow

Maker Initiates	Finance Officer (Maker) selects CSRApproved applications. Sets payment type (Direct/Institution), verifies amount, submits for approval.
Checker Reviews	Second Finance Officer (Checker) reviews payment batch. Cannot be same person as Maker.
Checker Approves	Payment status → PaymentInitiated. File generated with bank details and amounts.
Bank Transfer	Finance team executes transfer via their banking system. Uploads confirmation / transaction reference.
Status Updated	PaymentStatus → PaymentCompleted. ReferenceNo recorded. Receipt auto-generated.
Student Notified	Email + SMS to student: scholarship amount, bank transfer date, reference number.
Failure Handling	If transfer fails: status stays Initiated. Finance officer corrects details and retries. All retries logged in AuditLogs.

Chapter 9

Dashboard Design — 5 Role-Based Views

What each user role sees on their personalized dashboard

Student Dashboard

- Application Status Timeline — visual step indicator showing current stage
- Document Checklist — each doc: Pending / Uploaded / Verified / Rejected with reason
- Re-upload Prompt — flagged documents shown with rejection reason and deadline
- Matched Scholarships — scholarships auto-matched to their profile
- Messages & Clarifications — reviewer requests shown with response deadline
- Payment Status — once paid: amount, bank reference, date

Agent Dashboard

- My Students Summary — registered, submitted, approved, rejected counts
- Student List Table — all students registered via agent code with status
- Bulk Registration — upload CSV to register multiple students at once
- Approval Rate KPI — % of agent's students getting approved
- Commission Tracker — commission earned per approved scholarship
- Pending Follow-ups — students with incomplete applications needing action

Admin Console Dashboard

- Funnel Overview — total at each stage (registered → applied → audited → screened → approved → paid)
- Stage Bottlenecks — which stage has most applications stuck and for how long
- Eligibility Configuration — add/edit/delete scholarship rules without code
- CSR Partner Management — create shortlists, track partner response times
- Override Controls — can override any stage decision with logged reason
- SLA Violations — applications past SLA threshold highlighted in red
- Audit Log Viewer — searchable log of every action in the system

CSR Partner Dashboard

→ Fund Overview — Total provided, allocated, utilized, remaining (with progress bars)

→ Shortlist Review — students pending partner approval with eligibility summary

→ Beneficiary List — students approved and funded from their pool

→ Fund Distribution — breakdown by team, region, institution type (pie chart)

→ Disbursement Timeline — line chart of payments over time

→ Audit Export — download full audit report per beneficiary for CSR compliance

Finance Dashboard

→ Payment Queue — all CSRApproved applications ready for disbursement

→ Maker Actions — select students, set payment type, submit for checker review

→ Checker Review — approve or return payment batches

→ Payment History — all transactions with status, reference numbers

→ Failed Payments — failed transfers with retry option

→ Reconciliation Report — match payments to bank statements

UI Design — Screen by Screen

Visual layout and design system for every major screen

Design System

Element	Value	Reason
Primary Color	#5b2c6f (Purple)	Trust, education, professionalism
Accent	#2e86c1 (Blue)	CTAs, interactive, info links
Success	#0e6251 (Teal)	Verified, approved, completed
Warning	#f39c12 (Gold)	Pending review, SLA warning
Danger	#c0392b (Red)	Rejected, failed, overdue
Font	Inter / Geist Sans	Clean, readable at all sizes
Card Radius	8px	Professional, not childish
Sidebar	280px fixed	Room for labels + icons

Login Screen

→ Centered card (500px) on purple gradient. Logo + 'CSR Scholarship System' heading.

→ Email + Password. 'Student Register' secondary button. 'Agent Login' link.

→ After login: redirect to role-specific dashboard with personalized welcome.

Student — Multi-Step Application Form

→ Progress bar at top: Step 1 of 5. Can go back/forward freely.

→ Step 1 Personal: Name, DOB, Gender, Category, Address fields.

→ Step 2 Education: Course (autocomplete), Institution (from DB), Enrollment year.

→ Step 3 Family Income: Annual income field. Live eligibility preview shown based on rules.

→ Step 4 Bank Details: Account no + IFSC + re-enter confirmation. Auto-validates IFSC.

→ Step 5 Documents: Drag-drop upload area per document type. Progress per file.

→ Final: Full summary. 'Submit Application' button. Success screen with application ID.

Document Audit Screen (Reviewer)

→ Left: Application summary card — student name, applied date, scholarship, urgency badge.

→ Center: Checklist grid — each doc with status badge and action buttons.

→ Click doc: Full-screen preview overlay. Verify / Reject buttons below preview.

→ Reject: Text input for reason (required). Auto-email preview shown before confirming.

→ Progress bar at top: X of Y documents verified. Can only proceed when all done.

Background Check Screen

→ Table of check types required. Officer fills result per row (Pass/Fail/Inconclusive).

→ Evidence upload per check type. Notes field (required on Fail).

→ 'Submit Results' button only enabled when all check types have a result.

→ Fail path: Warning modal — 'This will route case for rejection decision. Confirm?'

CSR Partner — Approval Screen

→ Shortlist table: Student name, college, scholarship amount, eligibility score, doc status.

→ Batch actions: Select all → Approve All / Decline Selected.

→ Click student: Side panel with eligibility summary, college details, document checklist status.

→ No bank details visible (privacy). Only academic and eligibility info shown.

→ Comment box for each decision. Audit trail of all partner actions.

Separation of Duties & Compliance

How the system prevents same person from controlling multiple stages

Enforcement Rules

Core Principle: No single user can be the Doc Reviewer, Screening Officer, AND Finance Maker/Checker on the same application case. This is enforced at both the application logic layer and audited in AuditLogs.

Doc Reviewer vs Screener	When assigning a Screening Officer to a case, the system checks AssignedDocReviewer on that application and blocks the same user from being assigned as AssignedScreener.
Maker vs Checker	The Finance Checker who approves a payment batch cannot be the same user who initiated (Maker) that same batch. Enforced in payment submission API.
Admin Override	Admins can override any stage decision but MUST log a reason. Override actions are highlighted in AuditLogs and visible to CSR Partner in audit export.
CSR Partner vs Admin	CSR Partner sees aggregated data only. Cannot see individual student bank details. Cannot access admin configuration screens.
Audit Logging	Every role change, stage transition, approval, rejection, and override is written to AuditLogs with UserID, timestamp, old value, and new value.

SLA Reporting & Audit Export

Turnaround tracking, bottleneck identification, and CSR compliance export

SLA / Turnaround Report

The system tracks how long each application spends at every stage. This surfaces bottlenecks — which reviewer is slowest, which stage causes the most delays — enabling admin to redistribute workload.

Funnel Report	Registrations → Applied → AutoMatched → DocAuditComplete → BGCheckComplete → ScreeningApproved → CSRApproved → Funded. Shows drop-off at each stage.
Stage Turnaround	Per stage: avg days taken, min days, max days, SLA target (configurable), % within SLA. Broken down by reviewer, team, region.
Reviewer Performance	Per reviewer: applications handled, avg review time, approval rate, SLA compliance rate.
Bottleneck Alert	Flags: applications stuck at same stage for > SLA_THRESHOLD days. Admin receives daily digest.
Disbursement Reconciliation	All payments: application ID, student, amount, date, bank reference, payment type (Direct/Institution). Exportable for finance team.
Diversity Coverage	Beneficiaries by category (SC/ST/OBC/General), state, income band, institution type. Privacy-aggregated — no individual PII.

Audit Export for CSR Partner

What it contains (per beneficiary):

Student name, college, category, scholarship amount, payment date, payment reference

Eligibility evidence: which rules were met and proof (e.g. income certificate verified by officer X on date Y)

Document verification: each doc, verified by whom, on what date

Background check: result, officer, date

Screening decision: officer, date, notes

CSR partner approval: which partner representative, date

Format: Downloadable CSV + PDF report. Auto-generated per sponsor. Fully audit-ready for CSR reporting.

Notification System

When notifications fire, what they say, and how they are delivered

Trigger	When	Channel	Recipient	Message
Account Created	On registration	Email	Student	Welcome + login link + application guide
Application Submitted	On submit	Email + In-App	Student	Confirms submission. Shows application ID.
Doc Rejected	On doc rejection	Email + SMS	Student	Which doc, exact reason, re-upload deadline (7 days).
All Docs Verified	Audit complete	In-App	Student	Documents verified. Moving to background check.
BG Check Done	Check complete	In-App	Student	Verification done. Moving to screening.
Screening Approved	Approved	Email	Student	Shortlisted for scholarship. Awaiting sponsor approval.
Screening Rejected	Rejected	Email	Student	Application declined with reason.
CSR Partner Approved	Partner approves	Email + SMS	Student	Scholarship confirmed! Payment in process.
Payment Completed	After transfer	Email + SMS	Student	Amount, bank reference, date.
SLA Alert	Past SLA threshold	Email	Admin	X applications stuck at Y stage for Z days.
Shortlist Ready	Admin creates shortlist	Email	CSR Partner	Shortlist link with count of students pending approval.
New Assignment	Case assigned	Email + In-App	Reviewer/Officer	Application link, deadline, student summary.

Chapter 14

API Design Overview

REST endpoints organized by module — JWT auth required on all

Base URL: /api/v1/ | **Auth:** JWT Bearer Token | **Rate Limit:** 100 req/min per user | **Response:** { success, data, message }

Auth

POST	/auth/register	Register student or agent
POST	/auth/login	Login — returns JWT token
POST	/auth/forgot-password	Send OTP reset email/SMS

Students & Profile

GET	/students/me	Get own profile
PUT	/students/me	Update profile details
GET	/students/:id/eligibility	Check eligibility against all active scholarships

Applications

POST	/applications	Create draft application
POST	/applications/:id/submit	Submit for review
GET	/applications/:id	Get full application with all stage statuses
GET	/applications/my	Student's own applications
GET	/applications	Admin: list all with filters

Document Audit

POST	/documents/upload	Upload document to checklist item
PATCH	/documents/:id/verify	Mark document verified
PATCH	/documents/:id/reject	Reject with reason — triggers student notification

Background Check

POST	/bgchecks	Record background check result
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GET	/bgchecks/:applicationId	Get all checks for an application
-----	--------------------------	-----------------------------------

Screening & CSR

POST	/screening/:id/approve	Screening Officer approves application
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POST	/screening/:id/reject	Screening Officer rejects with reason
------	-----------------------	---------------------------------------

POST	/csr/shortlists	Admin creates shortlist for CSR partner
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POST	/csr/:id/approve	CSR Partner approves student
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POST	/csr/:id/decline	CSR Partner declines student
------	------------------	------------------------------

Disbursement

POST	/payments/initiate	Finance Maker submits payment batch
------	--------------------	-------------------------------------

POST	/payments/:id/approve	Finance Checker approves batch
------	-----------------------	--------------------------------

POST	/payments/:id/confirm	Upload bank confirmation + reference number
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GET	/payments	List all payments with filters
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Reports

GET	/reports/funnel	Stage-by-stage funnel counts
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GET	/reports/sla	SLA/turnaround per stage and reviewer
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GET	/reports/disbursement	Payment history and reconciliation
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GET	/reports/diversity	Coverage by category, region, income band
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GET	/reports/audit-export	Full audit report per beneficiary (sponsor use)
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Tech Stack & Deployment

Every tool used and why it was chosen

Layer	Technology	Why
Frontend	Next.js 14 + Tailwind CSS	SSR, fast, production-ready, mobile-responsive
UI Components	shadcn/ui + Lucide Icons	Professional, accessible, consistent design
Charts	Recharts + Chart.js	Flexible React-native charts for dashboards
Backend	Node.js + Express.js	Fast async runtime, huge ecosystem
Database	Microsoft SQL Server	Enterprise ACID, perfect for multi-role transactions
Query Builder	Knex.js	Full SQL control, SQL Server compatible
Auth	JWT + bcrypt	Stateless auth, secure password hashing
File Upload	Multer + AWS S3	Scalable secure document storage
Email	Nodemailer + SendGrid	Reliable delivery with HTML templates
SMS	Twilio / MSG91	India-friendly SMS gateway
Background Jobs	Bull Queue + Redis	Async auto-matching, notification queues
Validation	Joi + Zod	Input validation, type safety on all APIs
Monitoring	Sentry + Datadog	Error tracking, performance, uptime alerts
Deployment	Vercel (FE) + Render/AWS (BE)	Zero-config CI/CD, scalable cloud hosting

Document Summary

This document covers the complete technical design of the CSR Scholarship Management System — 12 SQL Server database tables, 8 distinct user roles with separation of duties enforcement, end-to-end lifecycle with auto-matching engine, document audit re-upload loop, background check module, CSR partner approval workflow, two-mode fund disbursement (student and institution), 5 role-based dashboards, SLA reporting with bottleneck detection, audit export for CSR compliance, 12-event notification system, 35+ REST API endpoints, and a modern Next.js + Node.js tech stack. Built to enterprise standards and fully audit-ready for CSR reporting requirements.